INTERNSHIP TIMELINE

Project Title: To predict Genetic Disease risk based on family history

Training Phase (Week 1-4)

Week	Week 01	Week 02	Week 03	Week 04
Focus Area	Introduction to AIML concepts and tools	Backend development using Flask and understanding API'S	Learning frontend technologies (HTML,CSS) and UI design	Data Handling, System Analysis and UI Enhancement

Project Development Phase (Week 5-16)

Week	Phase	Description
Week 05	Requirement Analysis	Identifying requirements for genetic disease prediction system and project objectives.
Week 06	Data preparation and Exploration	Creation and pre-processing of sample dataset along with missing values and outliers
Week 07	Feature Engineering	Integration of dataset upload and display of properties – Gene expressions and SNP markers
Week 08	System design	Designing system architecture, application and preparing system for ML integration
Week 09	UI Optimization	Enhancement of UI and user interaction mechanisms with form validation
Week 10	Theoretical Study	Study of ML algorithms and implementation of usability features
Week 11	Model Training and Implementation	Training and testing the machine learning model and evaluation using performance metrics
Week 12	Model Optimization	Hyperparameter tuning and addition of informative features
Week 13	System Refinement	Overall system refinement
Week 14	System Enhancement	Redesign of UI components and user experience
Week 15	Advanced Features	Improved validation and enhanced user mechanisms for improving user feedback
Week 16	Testing and Deployment	Final system testing and deployment on web platform with preparation of technical documentation